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1
Max σ will be hoop stress

$$\sigma_{\theta} = -\sigma^* (1 - 3\eta^2)$$

$$\sigma^* = \frac{k B \Delta T}{4(1-\nu)}$$

$$\Delta T = T_o - T_s$$

$$\sigma^* = \frac{10E-6 \cdot 200 \cdot \Delta T}{4(1-0.35)}$$

Need ΔT

$$\Delta T = \frac{LHR}{4\pi k} = \frac{300}{4 \cdot \pi \cdot 0.025} = 955 K$$

6/10

$$\sigma^* = \frac{10E-6 \cdot 200 \cdot 955 K}{4(1-0.35)} \rightarrow 734 \text{ MPa}$$

$$\sigma_{\theta, f} = 175 \text{ MPa}$$

- what is σ_{θ} ?
- where is it max?

$$175 \text{ MPa} = -734 \text{ MPa} (1 - 3\eta^2)$$

$$\frac{175}{734} = 1 - 3\eta^2$$

$$\eta = \sqrt{\left(1 - \frac{175}{734}\right)} = 0.873$$

= didn't divide by 3

∴ CRACK will extend 0.873% of WY into fuel

↓
this would be 87%

- does this make sense?

$$\frac{2}{p} = 15 \text{ MPa} \quad \bar{R}_e = 0.55 \text{ cm} \quad t_o = 0.05 \text{ cm}$$

$$a.) \sigma_\theta = \frac{pR}{\delta} = \frac{15 \cdot (0.55)}{0.05} = 165 \text{ MPa}$$

$$\sigma_z = \frac{pR}{2\delta} = 82.5 \text{ MPa}$$

$$\sigma_r = -\frac{p}{2} = -7.5 \text{ MPa}$$

$$R_i = 0.525$$

$$R_o = 0.575$$

$$\frac{R_o}{R_i} = 1.095$$

$$b.) \sigma_\theta = p \left[\frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} \right]$$

$$r = R_i$$

$$\sigma_\theta = 15 \left[\frac{(1.095)^2 + 1}{(1.095)^2 - 1} \right] = 165 \text{ MPa}$$

$$\sigma_z = \frac{15}{(1.095)^2 - 1} = 75.4 \text{ MPa}$$

$$\sigma_r = -15 \left[\frac{(1.095)^2 - 1}{(1.095)^2 - 1} \right] = -15 \text{ MPa}$$

3
First, need Refuse T profile

$$T_{co} - T_{ci} = \frac{LHR}{2\pi R_F} \frac{t_c}{K_c} = \frac{250}{2\pi \cdot .48} \cdot \frac{0.08}{0.15} = 44.2 K$$

$$T_s - T_i = \frac{LHR}{2\pi R_F} \frac{t_g}{K_g} = \frac{250}{2\pi \cdot .48} \cdot \frac{0.008}{0.003} = 221 K$$

13/16

$$T_o - T_s = \frac{LHR}{4\pi R_F} = \frac{250}{4\pi \cdot 0.03} = 663 K$$

$$\Delta \dot{Q}_g = \Delta R_c - \Delta R_F = K R_c \Delta T_c - K R_F \Delta T_F$$

$$K = 10 \cdot 10^{-6}$$

$$T_{co} = 550 K$$

$$\bar{T}_c = \frac{T_{ci} + T_{co}}{2}$$

$$\frac{550 + (550 + 44.2)}{2}$$

$$\bar{T}_c = 572.1$$

$$\bar{R}_c = \frac{0.48}{0.48 + \left(\frac{0.18}{2} + 0.48\right)} = 0.6$$

why 0.18?

→ too high, wrong geometry

$$\bar{T}_F \approx \frac{T_s + T_o}{2} = \frac{815 K + 1478}{2} = 1147 K$$

$$\Delta \dot{Q}_g = \Delta R_c - \Delta R_F = K R_c \Delta T_c - K R_F \Delta T_F$$

$$\Delta \dot{Q}_g = 10^{-6} \cdot 0.6 \cdot (572.1 - 300) - 145 \cdot 10^{-6} \cdot 0.48 \cdot (1147 - 300)$$

$$\Delta \dot{Q}_g = 0.001632 - 0.00056918$$

$$\Delta \dot{Q}_g = 4.05 \cdot 10^{-3} \text{ cm}$$

- gap increases? not right

4
 $\sigma_o = ? \quad \sigma_r = ? \quad r = R_i \quad R_i = 0.55 \quad t_c = 0.06$

$$\sigma_r = \frac{\Delta T}{2} \frac{KR}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{r} \left(\frac{r}{R_i} - 1 \right) \right)$$

5/6 $\Delta T = T_{\infty} - T_{ci} = 50 \text{ K}$

$$\sigma_r = \frac{50}{2} \cdot \frac{15 \cdot 10^{-6} \cdot 100 \cdot 10^6}{1 - 0.34} \left(\frac{r}{R_i} - 1 \right)$$

$$\therefore \sigma_r = 0, \text{ since } r = R_i$$

$$\sigma_o = \frac{\Delta T}{2} \frac{KR}{1-\nu} \left(1 - 2 \frac{R_i}{r} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_o = \frac{50}{2} \frac{15 \cdot 10^{-6} \cdot 100 \cdot 10^6}{1 - 0.34} \left(1 - 2 \frac{0.55}{0.06} \left(\frac{0.55}{0.55 - 0.06} - 1 \right) \right)$$

$$\sigma_o = 7.07 \text{ MPa}$$

~ too low
 ~ added a spurious term
 why t_c ?

5
 $\dot{F} = 6 \cdot 10^{13} \quad \Delta p_o = 0.015 \quad B_o = 5 \frac{\text{mWD}}{\text{kgU}} \quad p = 10.47$
 $t = 10 \text{ days}$

$$B_o = \frac{5}{0.50} = 0.00524 \text{ FIMA}$$

$$\epsilon_o = \Delta p_o \left(\exp \left(\frac{B h(0.01)}{c o B_o} \right) - 1 \right)$$

Need $B \rightarrow B = \frac{\dot{F} t}{N_u}$

$$N_u = 10.47 \frac{\text{g}}{\text{kg}} \frac{1 \text{ mol}}{270 \text{ g}} \frac{6.022 \cdot 10^{23}}{1 \text{ mol}}$$

$$N_u = 2.45 \cdot 10^{22}$$

$$B = \frac{6 \cdot 10^{13} \cdot (10 \text{ days} \cdot 24 \cdot 3600)}{2.45 \cdot 10^{22}} = 0.602116$$

$$\epsilon_o = 0.015 \cdot \left(\exp \left(\frac{0.002116 \cdot h(0.01)}{1 \cdot 0.00524} \right) - 1 \right)$$

$$\epsilon_o = -0.0126$$

4/4
6. elasticity → Ability to resist force & return to original shape
plasticity → After force is removed, non-reversible distortion/change in shape. Permanent deformation when force is removed
size when force is removed

3/8
7
Plasticity changes microstructure, usually increasing strength (σ_y)
Caused by stress/strain, but due to shifts in crystal lattice which "aligns" microstructure in more optimal way than before stress was present

- didn't really get it

3/3
8
Model temps in fuel, model temp in cladding, provide gap heat transport estimate. All three solve the fuel thermo/mech problem
stress

4/5
9
When sintering is done, particles are in different positions/orientations relative to one another, so grain boundaries form where things don't "line-up"
- needed more

6/8
10
Fuel pellets are not perfectly dense (see answer #a), and the microstructure impacts the material's thermal performance. This is critical in understanding how heat moves through fuel & cladding. For thermal conductivity, a microstructure-based model allows for a more accurate model which doesn't depend on burn-up, for example. Allows for less reliance on empirical models.
- needed more

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7/ microstructure → structure of material grains, has ^{close enough} impact on thermal performance. Can change ductility, temp behavior, etc.

8/ Processing → ex. Quenching → heating to temp for some time, then ^{rapidly cooled}
causes phase trans. w/out gradual stress removal.
[similar to ϵ/σ curves, increases σ_y] - didn't get this part

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2/4 ~~Whereas~~ At boundaries, intergranular diffusion is more significant than intragranular. (The grain is more ideal toward the center.) - huh?

13

6/ P_d drives → reduction in GB energy

P_r resists → caused by mobility inhibitors like pores, defects, etc.

14

6/ Densifies due to fuel ^{not being 100% dense} due to sintering process. Gaps in microstructure are "collapsed" as fuel continues to heat in operation. At micro-scale, occurs due to energy change associated with shrinking surface area / free E. ^{don't like this term}